

SymPy Bug Card

Bug 55: singularities Misses $\text{polylog}(1, x)$ Singularity

Task. Find the complex singularities of the order-one polylogarithm.

`singularities(polylog(1, x), x,  $\mathbb{C}$ )`

SymPy's answer.

`singularities(polylog(1, x), x,  $\mathbb{C}$ ) =  $\emptyset$ .`

Correct answer.

`1  $\in$  singularities(polylog(1, x), x,  $\mathbb{C}$ ).`

Explanation. The identity $\text{polylog}(1, z) = -\log(1 - z)$ gives $\text{polylog}(1, x) = -\log(1 - x)$, which has a logarithmic singularity at $x = 1$. SymPy itself evaluates both $\text{polylog}(1, 1)$ and $-\log(1 - 1)$ to `zoo`, so the empty set misses a concrete singular point.

Likely root cause. The implementation in `singularities` scans for a hard-coded set of visible singular forms such as powers and logarithms, but it does not handle `polylog` or expand $\text{polylog}(1, z)$ to $-\log(1 - z)$ before scanning. This is a new failure mode with no public precedent.

Artifact (GitHub).

[bug-report/bug-055-singularities-misses-polylog-1-x-singularity/](https://github.com/symengine/symengine.py/issues/1055)